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Roll no.

Degree: B.TECH Semester: 8th ECAM
END-SEMESTER EXAMINATION, APRIL-MAY, 2026

Course Title: Optimization Techniques in AI & ML

Course Code: EAEPE21

Duration: 03 Hours

Max. Marks: 40

Note: - Attempt all questions in the given order only. Missing data/information (if any), maybe suitably assumed & mentioned in the answer.

Q. No.	Question	Marks	CO
Q 1	Attempt any 2 parts of the following		
1a	Why is optimization required? How it can be implemented mathematically? Discuss with an engineering application.	4	CO1
1b	Is there any universal optimization algorithm? Justify your statement.	4	CO1
1c	What is the importance of an objective function and initial value in an optimization problem. Discuss with an example	4	CO1
Q 2	Attempt any 2 parts of the following		
2a	How are exploitation and exploration different in a metaheuristic approach? Explain with a real-world example.	4	CO2
2b	Initialize the parameters to minimize the function $X^2-10X+20$ using Simulated Annealing and compute for 2 iterations.	4	CO2
2c	Discuss with an example for each optimization algorithm as classified according to the principle of a biological and physical-based algorithms.	4	CO2
Q 3	Attempt any 2 parts of the following		
3a	Maximize $f(x) = x^2$ over $x \in [0,7]$ using an Adaptive Genetic Algorithm.	4	CO2,3
3b	In an ACO system, distances are: A-B = 2, A-C = 3. Initial pheromone $\tau = 1$, $\alpha = 1$, $\beta = 2$. Find probabilities from A to B and A to C.	4	CO3
3c	Discuss different variants of GA w.r.t. different crossover, mutation and selection methods.	4	CO3
Q 4	Attempt any 2 parts of the following		
4a	Describe the different variants of Particle Swarm Optimization (PSO). Discuss their key features and typical applications.	4	CO4
4b	Minimize the function $f(x) = x^2 + 7x + 10$, assuming five particles using two variants of PSO and comment upon the results obtained in terms of convergence, exploration and accuracy.	4	CO4
4c	Explain the concept of Fuzzy Particle Swarm Optimization (Fuzzy PSO). Describe how fuzzy rules are used to adapt PSO parameters.	4	CO4,5
Q 5	Attempt any 2 parts of the following		
5a	Explain the weighted sum average method for solving multi-objective optimization problems.	4	CO5
5b	What is Pareto optimal front? How are goals defined in a Multi-objective optimization problem of maximizing f_1 and minimizing f_2 ? Discuss with a suitable example.	4	CO5
5c	How NSGA-II can be used to solve multi-objective optimization problems?	4	CO5